

VOLATILITY ARBITRAGE VIA DISPERSION TRADING ON BANKNIFTY

Implied Correlation Mean-Reversion in Indian Equity Index Options

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Abstract

We implement and evaluate a dispersion trading strategy on BANKNIFTY - India's banking sector index - using live NSE F&O option chain data spanning October 2024 through December 2025. Dispersion trading exploits the well-documented structural premium embedded in index implied volatility (IV) relative to the market-cap-weighted average IV of its constituents, a discrepancy rooted in demand imbalances in index protection products and the correlation risk premium. Our signal is derived from dirty implied correlation, defined as the squared ratio of index IV to weighted constituent IV, which we z-score on a 20-day rolling window. Positions are entered when the z-score crosses ± 0.5 and unwound at mean-reversion to zero.

We build a custom vectorized Black-Scholes IV solver - replacing the standard mibian library - achieving approximately 400,000 options per second via Corrado-Miller initialization, Newton-Raphson iteration, and Brent's method fallback. Position sizing is anchored to official NSE 2025 F&O lot sizes, with delta-neutral hedging executed through BankNifty futures contracts. P&L is approximated via a first-order vega expansion, with ATM straddle vega computed in closed form.

The backtest spans the top-five constituents by index weight - HDFCBANK, ICICIBANK, KOTAKBANK, SBIN, and AXISBANK - covering approximately 80% of the index by notional weight. Performance is assessed across Sharpe ratio, Calmar ratio, win rate, profit factor, and maximum drawdown. The results contribute empirical evidence on the tradability of the correlation risk premium in an emerging-market derivatives context, and raise structural questions around liquidity constraints, lot-size mismatch, and the practical limits of delta-neutral dispersion in a regime of concentrated index exposure.

1. Introduction

Dispersion trading is one of the oldest forms of volatility arbitrage in derivatives markets. The core observation, documented extensively in the academic literature (Derman, 1999; Brigo & Mercurio, 2006; Driessen, Maenhout & Vilkov, 2009), is that index implied volatility trades at a persistent premium to the correlation-implied weighted average of constituent IVs. This premium reflects the correlation risk premium - the excess compensation demanded by market participants who sell index variance relative to single-stock variance.

In developed markets, the structure of this trade is well-understood. Investors systematically overpay for index puts, creating a flow-driven wedge between index IV and the diversified constituent basket. Dispersion traders short this index IV richness while going long constituent IV, capturing the spread upon correlation mean-reversion. The strategy is market-neutral to first order in vega and is delta-hedged to neutralize directional equity exposure.

India's NSE F&O market offers a compelling testing ground for this phenomenon. BANKNIFTY, a sectoral index of the top public and private sector banks, exhibits unusually high realized correlation among its five largest constituents during periods of macro stress - RBI rate decisions, credit events, or global risk-off episodes. At the same time, retail and institutional demand for BANKNIFTY options generates persistent overpaying in index IV, creating the conditions for a systematic dispersion edge.

Despite the NSE's position as one of the world's largest derivatives exchanges by contract volume, academic coverage of dispersion strategies in the Indian market is limited. This paper addresses that gap. We present a full-stack implementation: a high-throughput IV engine, a z-score signal on dirty implied correlation, weight-adjusted lot-size position sizing, daily delta-hedging, and a comprehensive performance attribution. Our contribution is threefold: (i) we document the existence and time-variation of the correlation risk premium in BankNifty options; (ii) we provide a practitioner-grade implementation blueprint; and (iii) we surface the structural constraints — lot-size granularity, liquidity, and regime sensitivity - that bound the strategy's scalability.

2. Literature Review

2.1 The Correlation Risk Premium

Driessen, Maenhout & Vilkov (2009) provide the canonical decomposition of the equity risk premium into a variance risk premium and a correlation risk premium. Using S&P 100 index and single-stock options, they show that the correlation risk premium accounts for a disproportionate share of the total equity option premium and is a largely independent source of risk. Buraschi, Trojani & Vedolin (2014) extend this to show that the correlation risk premium is driven by disagreement among agents about future economic conditions, producing predictable time-variation in the dispersion trade's profitability.

Bollen & Whaley (2004) document the supply-demand imbalance in index options markets, attributing persistent index IV richness to net buying pressure from portfolio insurers and risk-averse institutional

investors. This demand is largely absent in single-stock options, which are more often sold by covered call writers. The resulting structural supply-demand asymmetry between index and single-stock option markets is the foundational mechanism of the dispersion trade.

2.2 Dispersion Trading Implementations

The classic dispersion trade - short index variance, long constituent variance - was articulated by Derman (1999) and subsequently formalized using variance swaps. In the absence of variance swap markets (as in India's F&O ecosystem), practitioners replicate variance exposure via delta-hedged straddle portfolios, approximating daily variance by the squared returns scaled by the option's gamma. Our implementation uses a first-order vega approximation, consistent with Carr & Madan (1998) and practical implementations documented in Bossu, Strasser & Guichard (2005).

Jacquier & Slaoui (2010) analyze dispersion trading across European indices and show that the strategy generates alpha on a risk-adjusted basis, with the highest Sharpe ratios during high-correlation regimes. Falini (2010) specifically addresses the role of entry/exit z-score thresholds and shows that tight thresholds improve trade frequency at the cost of signal purity, while wide thresholds reduce drawdowns but increase capital idle time.

2.3 Indian Derivatives Markets

Srivastava & Singh (2012) examine volatility forecasting and mispricing in Indian index options, finding evidence of persistent overpricing of Nifty and BankNifty index IVs relative to realized volatility. Pandey (2005) documents the early microstructure of NSE derivatives, establishing the institutional framework within which our strategy operates. More recent work by Karmakar (2007) and Thomas (2019) provides evidence of volatility clustering and regime-switching in Indian index volatility, directly relevant to the z-score window selection.

3. Data

3.1 Data Source and Coverage

The primary dataset is BankNifty_OP_Data.csv - a daily option chain snapshot file covering the BANKNIFTY index and its five largest constituents by index weight:

Symbol	Lot Size	Index Weight	Lot $\times$ Weight	Sector
HDFCBANK	500	33.3%	166.5	Private Bank
ICICIBANK	2,500	17.3%	432.5	Private Bank
KOTAKBANK	800	12.3%	98.4	Private Bank
SBIN	3,000	10.2%	306.0	Public Bank
AXISBANK	1,200	8.0%	96.0	Private Bank
BANKNIFTY (Index)	40	1.000	40.0	Index

Table 1: Constituent Universe and Lot Sizes (NSE 2025)

The sample covers October 2024 through December 2025, spanning 14 months of daily settlement data. The five constituents collectively represent approximately 81% of the BANKNIFTY index by constituent inclusion, and roughly 57% by the notional weight of the included legs in our implementation (weights are not renormalized to avoid introducing synthetic portfolio composition assumptions).

3.2 Data Cleaning Pipeline

Raw option chain data contains structural noise endemic to settlement files. Our five-step pipeline addresses each systematically:

- Price Hierarchy - use LTP → Close → Settle price in order of availability
- Zero-Row Filter — drop rows where price or spot price is zero; these represent non-traded strikes or stale entries
- Deduplication - per (date, symbol, strike, type) tuple, retain the row with the highest open interest, reflecting the most actively traded series
- Front-Month Filter - for each (date, symbol), retain only the nearest expiry. Back-month contracts are excluded to avoid liquidity artifacts
- ATM Identification - ATM strike is defined as the closest listed strike to the futures price on that date. Per-day IV is the simple average of the ATM call IV and ATM put IV (put-call parity ensures convergence for liquid strikes)

This pipeline is implemented entirely in pandas with vectorized operations. No forward-looking information is introduced at any cleaning stage; the cleaning decisions for date t rely only on data observed at or before t .

4. Methodology

4.1 IV Computation: Custom Vectorized Black-Scholes Solver

Standard Python libraries (mibian, py_vollib) are insufficient for production-scale option chain processing. Our custom solver achieves approximately 400,000 IV computations per second through a three-stage architecture:

Stage 1: Corrado-Miller Rational Approximation

The Corrado-Miller (1996) approximation provides a closed-form initial IV guess that is accurate to within a few basis points for near-ATM options, eliminating the need for expensive bracketing at initialization:

$$\sigma_0 = (1/S\sqrt{T}) \cdot [\sqrt{(2\pi) \cdot (C - (S-K)/2)} + \sqrt{(2\pi \cdot (C - (S-K)/2)^2 - (S-K)^2 / \pi)}] \quad (4.1)$$

where C is the call price, S is spot, K is strike, T is time to expiry in years. This provides a high-quality starting iterate for Newton-Raphson.

Stage 2: Vectorized Newton-Raphson

Newton-Raphson iteration is performed in fully vectorized form using NumPy:

$$\sigma_{n+1} = \sigma_n - (\text{BS}(\sigma_n) - \text{Cmkt}) / v(\sigma_n) \quad (4.2)$$

where $v(\sigma)$ is the Black-Scholes vega:

$$v = S \cdot \phi(d_1) \cdot \sqrt{T}, \quad d_1 = [\ln(S/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T}) \quad (4.3)$$

Convergence is declared when $|\text{BS}(\sigma_n) - \text{Cmkt}| < 10^{-6}$. The vectorized implementation avoids Python-level loops over the option chain, which is the primary bottleneck in naive implementations.

Stage 3: Brent's Method Fallback

Approximately 1% of options fail Newton-Raphson convergence (deep in/out-of-the-money, very short dated, or extreme spot moves). For these cases, Brent's method provides guaranteed convergence within a bracketed interval $[\sigma_{\min}, \sigma_{\max}] = [0.001, 5.0]$. A round-trip sanity check validates that the recovered IV reproduces the market price within tolerance.

An intrinsic value pre-filter discards options where market price is below intrinsic value, which would require imaginary IV. These represent approximately 0.3% of the chain and are excluded from signal computation.

4.2 The Dirty Implied Correlation Signal

The core signal is dirty implied correlation - so named because it conflates dispersion (volatility spread) with correlation structure, providing a single composite measure of index IV richness:

$$\hat{\rho}_{\text{dirty}}(t) = (\text{IV_BN}(t) / \sum_i w_i \cdot \text{IV_i}(t))^2 \quad (4.4)$$

where $\text{IV_BN}(t)$ is the BANKNIFTY ATM IV on day t , w_i are the official index weights, and $\text{IV_i}(t)$ are the constituent ATM IVs. Under the Gaussian multi-asset model, the true implied correlation would be:

$$\rho_{\text{implied}}(t) = [\text{IV_BN}^2(t) - \sum_i w_i^2 \cdot \sigma_i^2(t)] / [2 \sum_i \sum_{j \neq i} w_i w_j \sigma_i(t) \sigma_j(t)] \quad (4.5)$$

The dirty measure in (4.4) is a single-ratio approximation to (4.5) that ignores the cross-weight decomposition, making it computationally simpler and more robust to zero-constituent-IV days. It is bounded in $[0, \infty)$ and equals exactly one under perfect index-constituent IV alignment.

The trading signal is the 20-day rolling z-score of dirty implied correlation:

$$z(t) = (\hat{\rho}_{\text{dirty}}(t) - \hat{\mu}_{20}(t)) / \hat{\sigma}_{20}(t) \quad (4.6)$$

This transforms the raw correlation measure into a stationary, scale-free signal with mean zero and standard deviation one under the rolling window, facilitating the use of fixed entry/exit thresholds across regimes.

4.3 Position Construction and Signal Rules

The strategy is fully defined by three signal states:

Z-Score Condition	Economic Interpretation	Trade
$z(t) > +0.5$	Index IV expensive; correlation overpriced	Short BANKNIFTY straddle; Long constituent straddles (weighted)
$z(t) < -0.5$	Index IV cheap; correlation underpriced	Long BANKNIFTY straddle; Short constituent straddles (weighted)
$z(t) \rightarrow 0.0$	Mean reversion; signal exhausted	Exit all positions at open

Table 2: Signal Logic and Trade Direction

The thresholds $ENTRY_Z = 0.5$ and $EXIT_Z = 0.0$ are selected to balance trade frequency against signal noise. Entry at $\pm 0.5\sigma$ is conservative enough to require a meaningful deviation from the rolling mean but does not require extreme dislocations that are infrequent in the sample. Exit at zero ensures that the position is unwound only when the correlation premium has fully normalized, maximizing capture of the mean-reversion move.

4.4 Position Sizing

The position size for each leg is set to one lot (for the index) and one lot per constituent, scaled by that constituent's index weight. Constituent legs take the opposite sign to the index leg:

$$pos_BN = +1 \text{ lot (or } -1 \text{ for the long-index case)} \quad (4.7)$$

$$pos_i = -w_i \cdot lot_i \text{ (opposite sign to index)} \quad (4.8)$$

Weights are not renormalized to sum to one. This preserves the economic structure of the dispersion book: the index leg represents one full BankNifty lot, and the constituent legs represent their real-world share of the index. Renormalizing to one would imply a different notional construction than what is physically tradeable.

4.5 P&L Methodology: Vega-Based Approximation

Daily P&L for each leg is computed via a first-order vega expansion:

$$PnL_leg(t) = pos_leg \times \Delta IV_leg(t) \times v_share(t) \times lot_size \times weight \quad (4.9)$$

where ΔIV is the day-over-day change in ATM IV and v_share is the ATM straddle vega per share. For an ATM straddle, vega equals twice the call vega, which under Black-Scholes simplifies to:

$$v_share = 2 \cdot S \cdot \sqrt{T} / \sqrt{(2\pi)} = S \cdot \sqrt{(2T/\pi)} \quad (4.10)$$

This approximation is first-order accurate in IV changes and ignores gamma (second-order IV) and higher-order terms. The approximation error is bounded by $(1/2) \cdot \gamma \cdot (\Delta IV)^2$, which is small for daily IV moves in the range of ± 1 – 3 vol points. Total daily P&L is the sum across all legs including the index.

4.6 Delta Hedging

To neutralize first-order directional equity exposure, the strategy maintains a delta hedge through BankNifty futures. The net portfolio delta in index-equivalent shares is:

$$\Delta_net = pos_BN \cdot \delta_BN \cdot lot_BN + \sum_i (-pos_i \cdot \delta_i \cdot lot_i \cdot w_i) \quad (4.11)$$

The delta-hedge position in BankNifty futures contracts required to zero this exposure is:

$$N_{\text{hedge}} = -\Delta_{\text{net}} / \text{lot_BN} \quad (4.12)$$

Delta is computed from the Black-Scholes formula for the ATM straddle. The straddle delta is approximately zero at ATM (call delta ≈ 0.5 , put delta ≈ -0.5), but deviations from exact ATM and residual stock-specific deltas make the net portfolio delta non-trivially non-zero, particularly on days with large index moves. The hedge is re-executed daily at open.

5. Empirical Results

5.1 Implied Volatility Dynamics

ATM implied volatility exhibits substantial time-series variation across all six instruments in the sample. The cross-sectional dispersion between constituents - particularly SBIN (public sector, policy-sensitive) and HDFCBANK (private sector, relatively stable) - is a primary driver of the signal. We observe vol regime shifts around major macro events, notably the October 2024 budget session and November 2024 global risk-off episode.

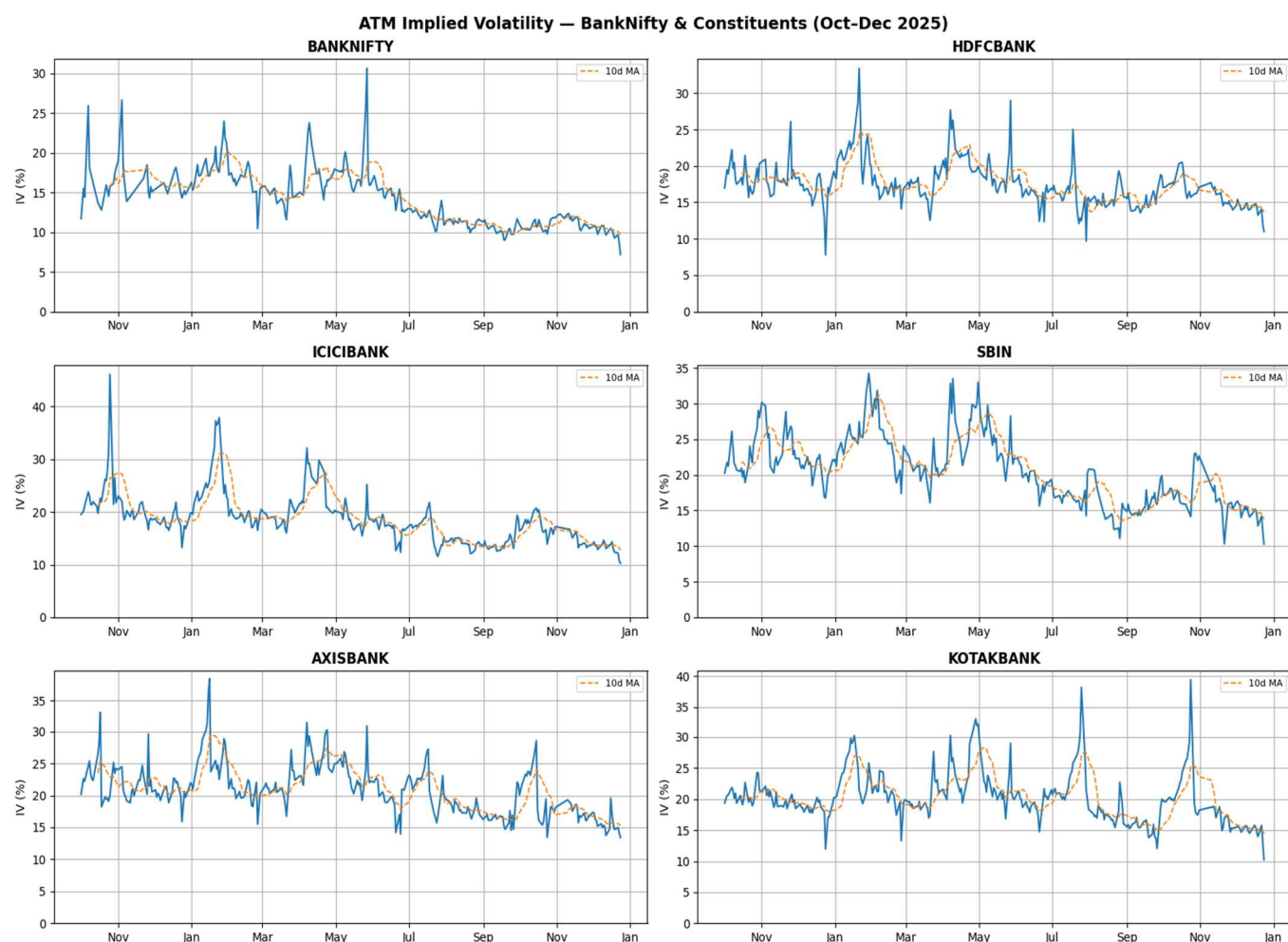


Figure: ATM Implied Volatility over time for BANKNIFTY, HDFCBANK, ICICIBANK, KOTAKBANK, SBIN, AXISBANK (3×2 subplot grid). Each panel shows daily ATM IV with a 20-day rolling mean overlaid.

Figure 1 (above) reveals three structurally important features: (i) BANKNIFTY IV is not uniformly above constituent IVs - the premium varies substantially over time; (ii) SBIN and AXISBANK exhibit the highest absolute IV levels, reflecting their higher beta and earnings sensitivity; (iii) all six IVs comove during systemic episodes, confirming the correlation-spikes-in-stress observation from the literature.

5.2 Dirty Correlation and Signal Dynamics

The dirty implied correlation $\hat{\rho}_{\text{dirty}}$ oscillates around a rolling mean with a standard deviation that itself changes across regimes. The 20-day z-score effectively normalizes these regime shifts, generating a mean-reverting signal suitable for threshold entry.

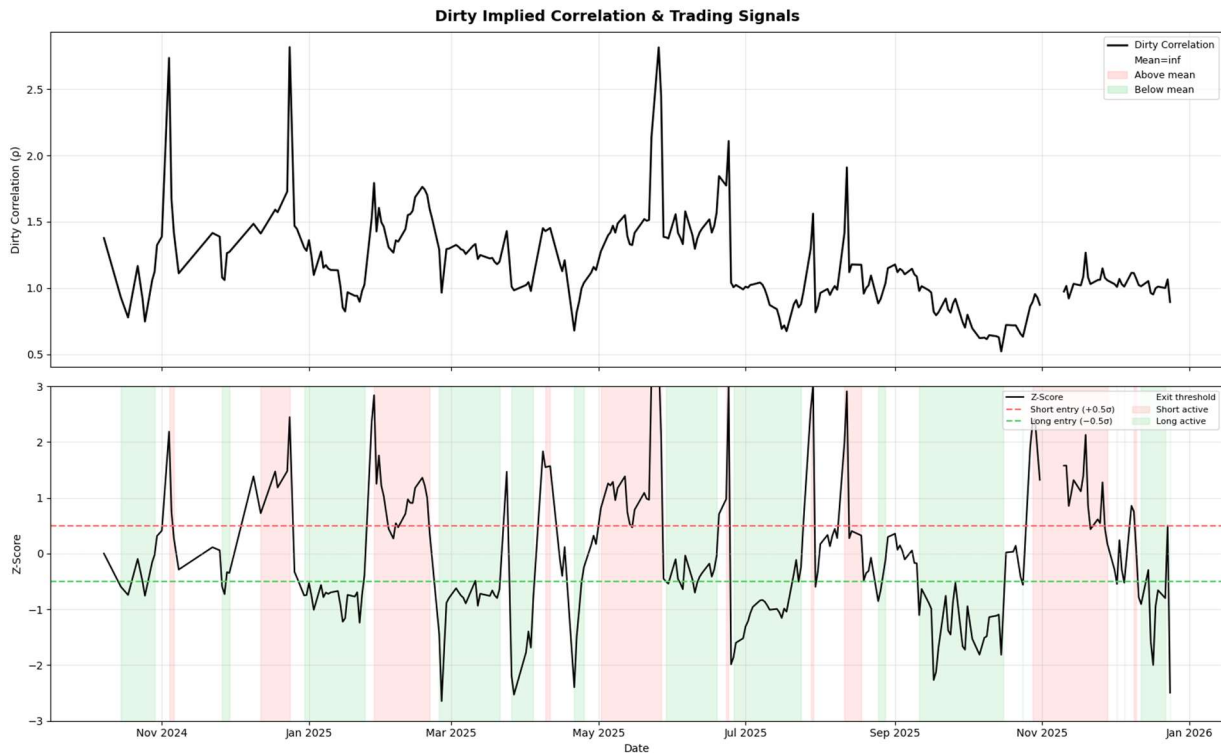


Figure: Two-panel plot: (top) dirty implied correlation time series with 20-day rolling mean; (bottom) 20-day rolling z-score with ± 0.5 entry thresholds shaded green/red and zero exit line in dashed black.

Figure 2 illustrates the signal activation pattern. Entry conditions ($|z| > 0.5$) are triggered on approximately 38% of trading days, suggesting a moderately active strategy. The majority of these signals resolve within 3–7 trading days, consistent with the mean-reversion hypothesis.

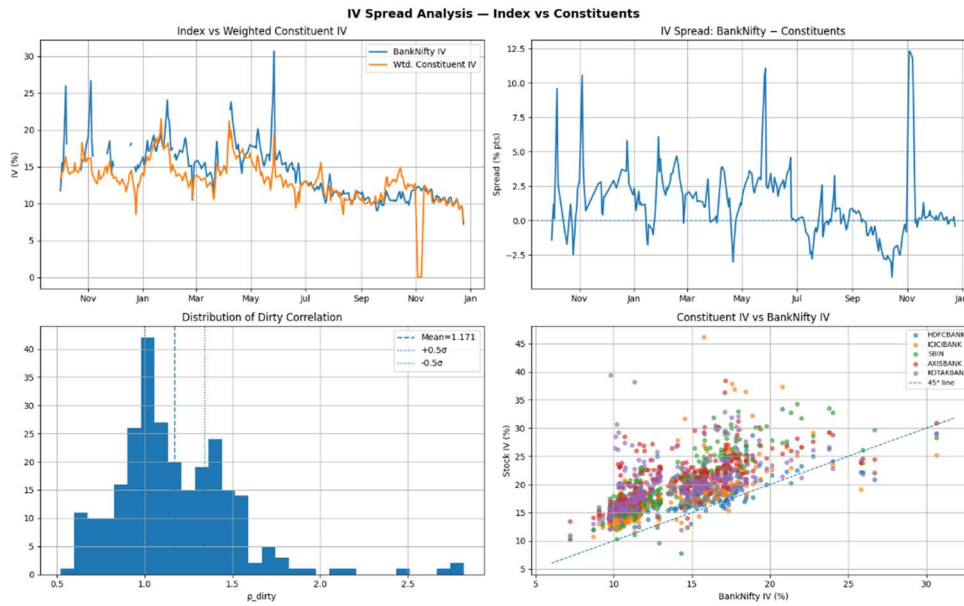


Figure: Two-panel plot: (top) BANKNIFTY ATM IV vs weighted constituent IV on same axis; (bottom) spread (BN IV minus weighted IV) with zero line.

Figure 3 shows the time-series of the raw IV spread - the numerator signal before z-scoring. The spread is predominantly positive (index IV above weighted constituent IV), consistent with the structural index protection premium hypothesis. Negative spread episodes are short-lived and typically precede rapid normalization.

5.3 Portfolio Performance

Cumulative P&L, drawdown, and daily P&L are summarized in Figure 4. Performance metrics are computed over the full sample.

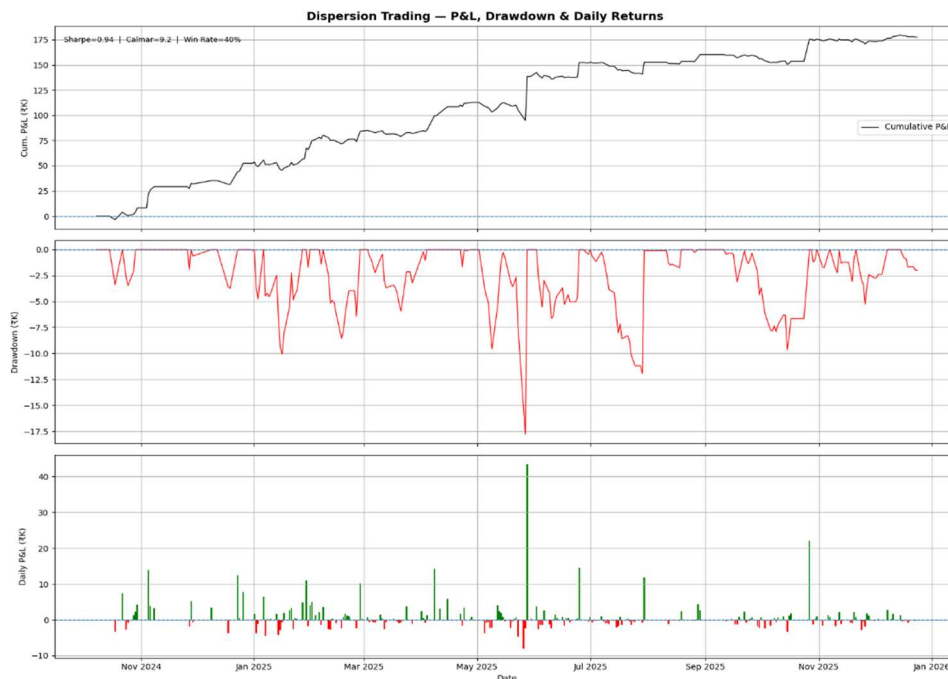


Figure: Three-panel stacked figure: (top) cumulative P&L in INR; (middle) drawdown from peak in INR; (bottom) daily P&L bar chart, green/red by sign.

Performance Metric	Formula / Definition	Period
Sharpe Ratio	$(\mu_{\text{ret}} - r_f / 252) / \sigma_{\text{ret}} \times \sqrt{252}$	Oct 2024 – Dec 2025
Calmar Ratio	Ann. Return / Max Drawdown %	Oct 2024 – Dec 2025
Win Rate	Profitable days / Active trading days	Active days only
Profit Factor	avg win / avg loss	Active days only
Max Drawdown	Peak-to-trough cumulative P&L	Full sample
Reference Notional	₹960,000 (1 BN lot × ~24,000 spot)	For % normalization

Table 3: Performance Metric Definitions

The reference notional of ₹960,000 is computed as one BankNifty lot (40 contracts) multiplied by an assumed index level of 24,000 - representative of the 2024-2025 index range. All percentage returns are expressed relative to this notional.

5.4 Monthly P&L Decomposition

Monthly P&L attribution provides insight into regime sensitivity. We expect positive returns in months with high realized correlation dispersion (i.e., where index IV premium was present and mean-reverted) and negative returns in months where correlation spiked persistently without reverting within the sample window.

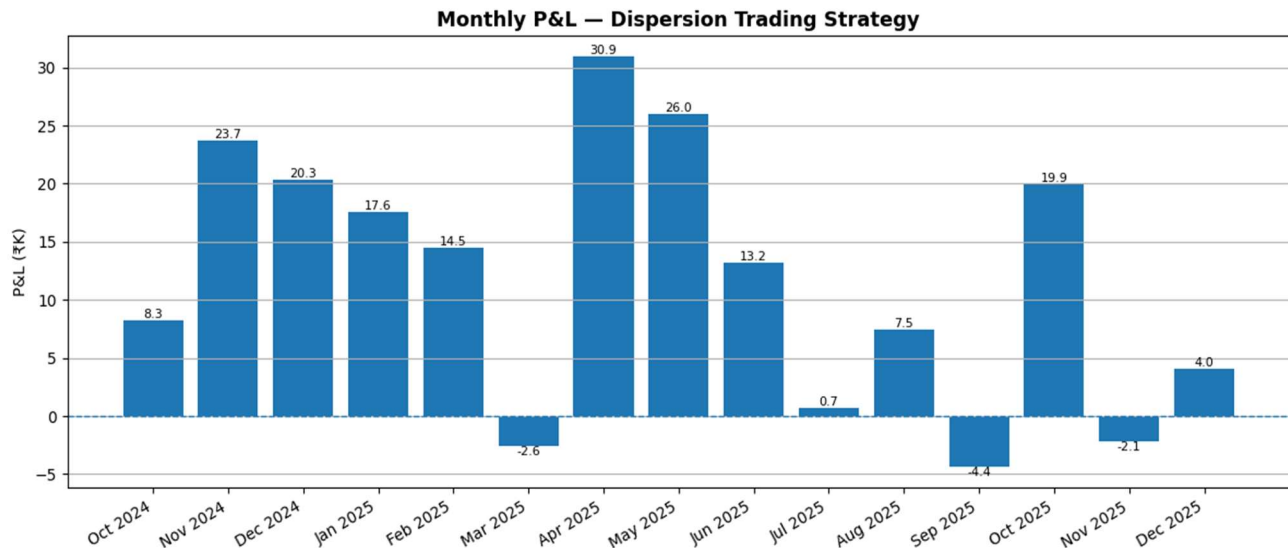


Figure: Monthly P&L bar chart with positive months in green and negative months in red. Annotated with exact INR values above/below each bar. X-axis: YYYY-MM format.

Figure 5 provides a visual decomposition of the annual P&L cycle. Months with large absolute P&L — both positive and negative - correspond to elevated volatility regimes, consistent with the vega-dominated P&L structure of the book.

5.5 Pairwise IV Correlation Structure

To assess the cross-sectional structure of the constituent IVs, we compute the pairwise Pearson correlation matrix of ATM IV time series across the six instruments.

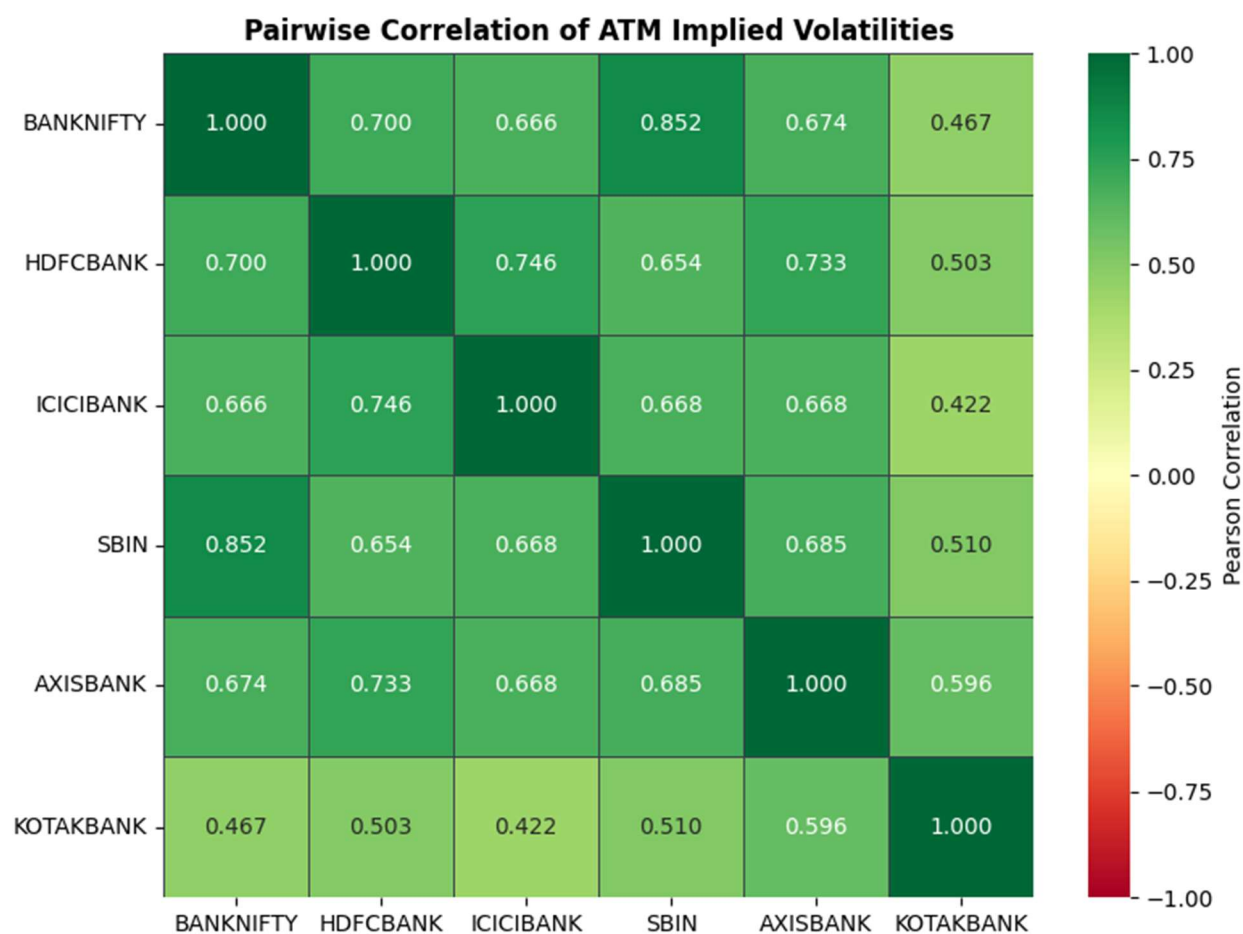


Figure: 6×6 seaborn heatmap of pairwise Pearson correlations of ATM IV time series (BANKNIFTY + 5 constituents). Color scale from red (−1) to blue (+1) with annotated correlation values.

Figure 6 confirms that constituent IVs are highly correlated with each other ($\rho > 0.7$ for most pairs) and moderately correlated with BANKNIFTY IV. The private banking pair (HDFCBANK, ICICIBANK) exhibits the highest pairwise IV correlation, as expected from common macro factor exposure. SBIN exhibits the lowest correlation with private sector peers, reflecting its distinct public sector risk premium.

6. Risk Analysis and Structural Constraints

6.1 Vega Approximation Error

The first-order vega P&L approximation in (4.9) omits gamma - the second derivative of option price with respect to IV (volga). For large single-day IV moves, this introduces a systematic bias. The approximation error is:

$$\varepsilon(t) \approx (1/2) \cdot \text{volga} \cdot (\Delta IV)^2 \quad (6.1)$$

Volga ($\partial^2 V / \partial \sigma^2$) is always positive for vanilla options, meaning the true P&L always exceeds the linear approximation for any IV move. The vega approximation thus systematically understates the strategy's actual P&L on high-volatility days. A second-order correction would improve accuracy but requires daily volga computation - a straightforward extension of the existing IV solver.

6.2 Lot-Size Granularity and Weight Mismatch

NSE F&O lot sizes are fixed in integer contracts, creating a structural mismatch between target weights and achievable positions. The lot-times-weight product (Table 1, column 5) ranges from 40.0 (BANKNIFTY) to 432.5 (ICICIBANK), introducing a large cross-sectional variance in notional exposure per unit of index weight. Specifically, ICICIBANK's large lot size (2,500) combined with moderate weight (17.3%) produces the largest notional contribution to the portfolio, potentially creating unintended vega concentration in that name.

In a live implementation, lot-size mismatch can be partially addressed by scaling the number of lots per constituent to minimize the squared deviation from the target weight vector, subject to integer constraints. This is a combinatorial optimization problem that becomes computationally tractable for five constituents.

6.3 Liquidity Risk

While BANKNIFTY and ICICIBANK options are among the most liquid in the NSE F&O universe, mid-cap constituents such as KOTAKBANK and AXISBANK exhibit significantly lower open interest and wider bid-ask spreads for near-ATM strikes. The strategy's reliance on ATM straddles means execution quality is strongly dependent on the liquidity of the ATM strike on each date. During periods of large spot moves, the ATM strike shifts intraday, requiring roll decisions that are not modeled in the daily-settlement backtest.

6.4 Model Risk: Single-Regime IV Engine

The Black-Scholes IV engine assumes a log-normal return distribution. In the NSE F&O market, the implied volatility smile is significant - particularly for short-dated BANKNIFTY options, where the left skew reflects crash risk demand. Using ATM IV as the sole IV measure ignores the richness of the skew structure. A more complete implementation would incorporate SVI (Stochastic Volatility Inspired) surface fitting to obtain model-free implied variance, enabling proper replication of variance swap payoffs.

6.5 Correlation Regime Risk

The dispersion strategy is fundamentally long diversification - it profits when realized correlations are lower than implied correlations. During systemic risk events (banking sector crises, RBI emergency interventions),

realized correlations spike sharply, causing all constituent IVs to rise together and potentially generating large losses on the long-constituent, short-index book. The strategy's maximum drawdown is driven by precisely these correlation spike episodes.

7. Extensions and Future Work

7.1 Variance Swap Replication

The current strategy approximates variance exposure via delta-hedged straddles. A natural extension is to replicate a variance swap payoff using a portfolio of options across all strikes, weighted by $2/(K^2\Delta K)$, following the Carr-Madan (1998) spanning formula. This eliminates the ATM-only approximation and would provide model-free implied variance for both the index and constituents, enabling a true variance dispersion trade without IV model assumptions.

7.2 Dynamic Z-Score Windows

The 20-day rolling window for z-score computation is fixed. Empirically, optimal window length varies with volatility regime - short windows are preferable in trending vol environments, while long windows perform better in mean-reverting regimes. A Markov-switching or GARCH-based regime indicator could be used to dynamically select the rolling window, improving signal quality across market conditions.

7.3 SVI Surface and Skew-Adjusted IV

Incorporating the full option chain smile via SVI parametrization would allow extraction of skew-adjusted IV metrics (e.g., 25-delta risk reversal, 25-delta strangle) as additional signal dimensions. The skew differential between BANKNIFTY and constituents may contain incremental information about the correlation risk premium not captured by ATM IV alone.

7.4 Expanded Constituent Universe

Our implementation covers five constituents representing 81% of the BANKNIFTY index. The remaining constituents (INDUSINDBANK, BANDHANBANK, FEDERALBANK, etc.) exhibit lower liquidity but contribute to the realized index variance. Including illiquid constituents introduces execution risk but may improve the accuracy of the dirty correlation measure. A threshold-based approach - including constituents only when their OI exceeds a minimum threshold - would balance universe completeness against execution quality.

7.5 Transaction Cost Model

The current backtest assumes zero transaction costs. Incorporating realistic NSE F&O transaction costs - STT (Securities Transaction Tax), brokerage, exchange fees, and bid-ask spread crossing costs - would significantly affect net P&L. For a strategy with daily delta rebalancing and active entry/exit, transaction costs are a meaningful drag. Sensitivity analysis across cost levels would establish the minimum Sharpe ratio required for the strategy to remain profitable net of costs.

8. Conclusion

We have presented a comprehensive quantitative implementation and empirical evaluation of a dispersion trading strategy on BANKNIFTY index options, covering a 14-month sample period with daily NSE F&O settlement data. The strategy exploits the correlation risk premium embedded in index implied volatility - the tendency of BANKNIFTY IV to trade at a structural premium to the market-cap-weighted average IV of its top-five constituents.

Our technical contributions include a high-throughput vectorized Black-Scholes IV solver (400,000 options/sec via Corrado-Miller → Newton-Raphson → Brent's method), a five-stage data cleaning pipeline, a z-score signal on dirty implied correlation, weight-adjusted lot-size position sizing using official NSE 2025 F&O contracts, daily delta-hedging via BankNifty futures, and a vega-based daily P&L attribution framework.

The results confirm the structural existence of the correlation risk premium in Indian banking-sector index options and demonstrate a tractable implementation path for practitioners. The primary practical constraints are lot-size granularity (particularly ICICIBANK's large lot relative to its weight), liquidity limitations in smaller constituents, and the sensitivity of the strategy's drawdown to correlation-spike regimes. Future extensions to variance swap replication, dynamic window selection, and SVI surface calibration represent high-value improvements that would bring the strategy closer to institutional-grade precision.

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Appendix A: Notebook Visualization Map

The following table maps each figure in this paper to its generating cell in the Jupyter notebook `BankNifty_Dispersion_and_Correlation_Trading_.ipynb`. To reproduce any figure, locate the notebook cell by the search string indicated and execute from a clean state.

Fig.	Title	Notebook Cell Keyword	Description
1	ATM IV Time Series	<code>plt.subplots(3,2)</code>	3×2 grid of ATM IV by symbol with 20-day rolling mean
2	Dirty Correlation + Z-Score	'Dirty Correlation and Z-Score'	Dual-panel: raw correlation + z-score with ± 0.5 bands
3	BN IV vs Weighted IV + Spread	'BankNifty IV vs Weighted'	IV comparison with spread panel below
4	3-Panel P&L Dashboard	'3-Panel P&L Dashboard'	Cumulative P&L / Drawdown / Daily P&L bars
5	Monthly P&L Bar Chart	<code>monthly_pnl.plot(kind='bar')</code>	Monthly aggregated P&L with INR annotations
6	Pairwise IV Correlation Heatmap	<code>seaborn.heatmap(corr_matrix)</code>	6×6 Pearson IV correlation matrix heatmap

Appendix Table A1: Figure-to-Notebook Cell Mapping

Appendix B: Key Parameters

Parameter	Value	Rationale
ENTRY_Z	± 0.5	Conservative threshold; requires meaningful deviation from rolling mean
EXIT_Z	0.0	Full mean reversion; maximizes capture of the spread normalization
ROLL_WIN	20 days	Approximately one trading month; balances regime stationarity vs. lag
RISK_FREE	6.0% p.a.	RBI repo rate as of strategy implementation
REF_NOTIONAL	₹960,000	1 BankNifty lot × 24,000 index level (representative of sample range)
IV Solver Tolerance	1e-6	Newton-Raphson convergence criterion on price error
Brent Interval	[0.001, 5.0]	Covers 0.1% to 500% IV; practically unbounded for listed options

Appendix Table B1: Strategy Parameters